

Roll No. 24 GMITCS103

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. III Semester I midterm Exam 2025-26
3CSI-02 Managerial Economics and Financial Accounting
Branch-CSE/ME/CE/EE
28-10-25/Tuesday

Time: 2 Hours**Maximum Marks: 40****Attempt all questions**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. no supplementary sheet shall be issued in any case.

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1 & CO2(Part-A)	6	1	0	2	1	10	10
CO1 & CO2(Part-B)	0	3	1	1	0	5	15
CO1 & CO2(Part-C)	0	0	1	1	1	3	15

Part A (Answer should be given up to 25 words only)
All questions are compulsory

- | | |
|---|----------|
| Q.1 Define economics. | (CO1) L1 |
| Q.2 Give one example each of a real flow and a monetary flow in an economy. | (CO2) L1 |
| Q.3 What do you understand by national income? | (CO1) L2 |
| Q.4 What is the difference between GDP and GNP? | (CO1) L4 |
| Q.5 What is individual demand? | (CO2) L1 |
| Q.6 What is demand forecasting? | (CO2) L1 |
| Q.7 Name two determinants of supply. | (CO2) L1 |
| Q.8 State two differences between deductive and inductive methods. | (CO1) L4 |
| Q.9 What is the law of supply? | (CO2) L1 |
| Q.10 Explain the circular flow of income in a two-sector economy. | (CO1) L5 |

10x1 = 10

Part B Analytical/Problem solving questions
Attempt all questions (word Limit 75)

- | | |
|---|----------|
| Q.1 "Economics is the study of scarcity and choice." Justify this statement with an example. | (CO1) L2 |
| Q.2 Explain how the concept of elasticity of demand can help a business in deciding pricing strategy. | (CO2) L2 |
| Q.3 Explain the determinants of demand. How do changes in these factors affect the demand curve? | (CO2) L3 |
| Q.4 Discuss any two challenges faced in the estimation of national income in a developing country like India. | (CO1) L2 |
| Q.5 Compare positive economics and normative economics with examples. | (CO1) L4 |

5 x 3 = 15

Part C (Descriptive/Analytical/Problem Solving/Design Question)
Attempt all questions (word Limit 125)

- | | |
|---|----------|
| Q.1 What is demand forecasting? Explain methods used in forecasting future demand. | (CO2) L3 |
| Q.2. "Economics is both a science and an art." Explain this statement and discuss the nature and scope of economics in solving real-world issues. | (CO1) L4 |
| Q.3 Define national income and explain the three main methods of measuring it. What are the difficulties faced in the measurement of national income in developing countries? | (CO1) L5 |

3 x 5 = 15

Roll No. 2466JCS103

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. III Semester II midterm Exam 2025-26
3CS1-02 Managerial Economics and Financial Accounting
Branch-CSE/ME/CE/EE
17-12-25/WEDNESDAY

Time: 2 Hours**Maximum Marks: 40****Attempt all questions**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. no supplementary sheet shall be issued in any case.

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1 (Part-A, B, C)	3	2	0	0	0	5	13
CO2 (Part-A, B, C)	0	2	0	1	0	3	5
CO3 (Part-A, B, C)	1	1	1	0	0	3	7
CO4 (Part-A, B, C)	2	1	0	0	1	4	10
CO5 (Part-A, B, C)	0	3	0	0	0	3	5

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

- | | |
|--|---------|
| Q.1 Define Managerial Economics. | (CO1)L1 |
| Q.2 Define National Income. | (CO1)L1 |
| Q.3 What do you mean by Law of Demand? | (CO2)L2 |
| Q.4 Define price elasticity of demand. | (CO2)L2 |
| Q.5 Define Production Function. | (CO3)L2 |
| Q.6 What is opportunity cost? | (CO4)L1 |
| Q.7 What do you mean by Monopoly? | (CO4)L1 |
| Q.8 Define Financial Statement Analysis. | (CO5)L2 |
| Q.9 What is Pay Back Period? | (CO3)L1 |
| Q.10 Explain Debtors Turnover Ratio. | (CO5)L2 |
- 10x1 = 10**

Part B Analytical/Problem solving questions**Attempt all questions (word Limit 75)**

- | | |
|--|---------|
| Q.1 Distinguish between deductive and inductive methods in Economics. | (CO1)L1 |
| Q.2 Discuss the various concepts of national income – Gross National Products, Net National Products, Personal Income and Disposable Income. | (CO1)L2 |
| Q.3 Distinguish between monopolistic competition and perfect competition. | (CO4)L2 |
| Q.4 Explain the degrees of price elasticity of demand. | (CO2)L4 |
| Q.5 Define Balance Sheet. Give two characteristics of balance sheet. | (CO5)L2 |

5 x 3 = 15**Part C (Descriptive/Analytical/Problem Solving/Design Question)****Attempt all questions (word Limit 125)**

- | | |
|---|---------|
| Q.1 Discuss the nature and scope of Managerial Economics. | (CO1)L2 |
| Q.2 Explain the Law of Variable Proportions. Explain various stages of this law with the help of a diagram. | (CO3)L3 |
| Q.3 How the price and output is determined under perfect competition during short period? | (CO4)L5 |

3 x 5 = 15

Roll No. 24E67CS103

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. III Semester I Midterm Exam 2025-26
3CS4-06/ Object Oriented Programming
CSE/IT/AI&DS/IOT/Cyber Security
29-10-2025 / Wednesday

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Part A (Answer should be given up to 25 words only)
All questions are compulsory

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1(Part-A,B,C)	4	3	1	2	1	11	19
CO2(Part-A,B,C)	1	1	3	1	1	7	21

- Q.1 Describe the default access specifier for members of a class in C++? (CO1) L1
Q.2 Explain the concept of class with syntax. (CO1) L2
Q.3 List the name of four programming paradigm. (CO1) L1
Q.4 Define Object with example? (CO1) L1
Q.5 Explain the scope of variable in c++. (CO1) L2
Q.6 In C++, which symbol is used to declare a reference variable? (CO2) L1
Q.7 State the operators used for dynamic memory allocation and deallocation in C++? (CO1) L2
Q.8 Can a function in C++ have more than one default argument? (Yes/No). (CO1) L4
Q.9 State the purpose of this pointer in C++? (CO2) L2
Q.10 Can a friend function access private data members of a class and Which keyword is used to declare a friend function inside a class?

10x1 = 10

Part B Analytical/Problem solving questions
Attempt all questions (word Limit 75)

- Q.1 Explain function overloading and its types? (CO2) L3
Q.2 Describe friend function? Also state the benefits of friend function. (CO2) L3
Q.3 Express structure and differentiate between class and structure in C++. (CO1) L4
Q.4 Create a Calculator Class having two data members and four methods like setValue, Add, Sub, Print. (CO2) L3
Q.5 Explain all Access Specifiers in OOPs. (CO1) L5

5 x 3 = 15

Part C (Descriptive/Analytical/Problem Solving/Design Question)
Attempt all questions (word Limit 125)

- Q.1 Describe concepts, advantages and applications of Object-Oriented Programming (OOP) with examples. (CO1) L3
Q.2 Define a constructor and a destructor. Give their syntax and explain it with example. (CO2) L5
Q.3 State and explain inline function with its advantages and disadvantages. (CO2) L4

3x 5 = 15



GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. III Semester II Midterm Exam 2025-26
3CS4-06/ Object Oriented Programming
Branch-CS/IT/AIDS/IOT/Cyber Security
 18/12/2025/Thursday

Roll No. 24607CS102

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Part A (Answer should be given up to 25 words only)

All questions are compulsory

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1(Part-A,B,C)	3	0	0	1	0	4	10
CO2(Part-A,B,C)	1	2	1	0	1	5	13
CO3(Part-A,B,C)	1	0	3	0	0	4	10
CO4(Part-A,B,C)	0	1	0	2	0	3	5
CO5(Part-A,B,C)	0	0	2	0	0	2	2

- Q.1 Write any four characteristics of procedure oriented programming.
 Q.2 Define the different access specifiers.
 Q.3 What are inline functions?
 Q.4 What is function overloading?
 Q.5 What is abstract class?
 Q.6 What is function overriding?
 Q.7 What is operator overloading?
 Q.8 What is dynamic binding?
 Q.9 What are exceptions?
 Q.10 What are templates?

(CO1) L1
 (CO1) L1
 (CO2) L1
 (CO2) L2
 (CO3) L1
 (CO3) L3
 (CO4) L2
 (CO4) L4
 (CO5) L3
 (CO5) L3

10x1 = 10

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

- Q.1 Explain the characteristics of object oriented programming.
 Q.2 Explain the use of new and delete operators with suitable examples.
 Q.3 Explain the concept of virtual base class with suitable example.
 Q.4 What are static data members and member function? Write its any two characteristics of each.
 Q.5 What is this pointer? Explain its any two uses with suitable examples.

(CO1) L1
 (CO2) L2
 (CO3) L3
 (CO4) L4
 (CO2) L5

5 x 3 =15

Part C(Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

- Q.1 Write a program to create a class "employee" consist of name, age and salary and member functions for input/output. Enter the records of five faculty and seven staff and print them using array of objects. (CO1) L4
 Q.2 What are constructors? Write its any five characteristics. Explain different types of constructors with suitable examples. (CO2) L3
 Q.3 What is inheritance? Explain different types of inheritance with suitable example. (CO3) L3

3x 5 =15

Roll No. 24F6JCS703**GLOBAL INSTITUTE OF TECHNOLOGY**

B. Tech. III Semester II-Midterm Exam 2025-26

3CS2-01/3AID2-01/3CS2-01/3CIT2-01/3CCS2-01

Branch-CS/AIDS/IT/IOT/CYBER SECURITY

Subject: Advanced Engineering Mathematics

18/12/2025 Thursday

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. no supplementary sheet shall be issued in any case.

MID-TERM-2

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1(Part-A, B, C)	1				1	2	6
CO2 (Part-A,B,C)	1	1		1		3	5
CO3(Part-A, B, C)		2		1		3	5
CO4(Part-A, B, C)	1		2	1		4	8
CO5(Part-A, B,C)	2	1	1	1	1	6	16

Part A (Answer should be given up to 25 words only)

All questions are compulsory

10x1 = 10

- Q.1 What is Lagrangian function? (CO4) L1
 Q.2 What is optimization? (CO3) L2
 Q.3 What is the difference between liner and non-liner programming? (CO3) L2
 Q.4 What is the difference between slack variable and surplus? (CO5) L1
 Q.5 Define the Binomial distribution and write its mean and variance? (CO2) L2
 Q.6 If $E(X)=4$ and $E(Y)=1$ then what is the value of $E(2X+3Y)$? (CO1) L1
 Q.7 How many numbers of normal equations are required for fitting polynomial of m degree by least square method? (CO2) L1
 Q.8 If the given LPP has an optimal solution, then what about the dual solution? (CO5) L3
 Q.9 For non-degenerate feasible solution of $m \times n$ transportation problem, how many independent individual positive assignments will be required? (CO5) L1
 Q.10 Under what circumstance can the conditions $f'(x) = 0$ not be used to find the minimum of the function? (CO4) L3

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

5x3 = 15

Q.1. discuss the rank correlation coefficient for the data given below

X	10	12	15	14	19
Y	40	41	48	60	50

Q.2 Find the Maxima and Minima of the function $u = x^3 + y^3 - 3x - 12y + 25$.

(CO2) L4

(CO4) L3

Q.3. Using the Lagrange's multiplier method, solve the following problem-

$$Z_{Max} = 4x_1 - x_2 + 8x_2 - x_{22}$$

$$\text{Sub. to } x_1 + x_2 = 4$$

$$\text{and } x_1, x_2 \geq 0$$

$$4x_1 - x_2 + 8x_2 - x_{22}$$

(CO4) L4

Q.4 Construct the dual of the following problem-

$$Z_{min} = x_1 - 3x_2 + 3x_3$$

$$\text{Sub. to } 3x_1 - x_2 + 2x_3 \leq 7$$

$$2x_1 - 4x_2 \leq 12$$

$$4x_1 - 3x_2 - 8x_3 \geq 10$$

$$\text{and } x_1, x_2, x_3 \geq 0$$

(CO5) L2

(CO3) L4

Q.5 Write fifteen engineering applications of optimization techniques.

Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

3x5 = 15

Q.1 Solve the following transportation problem by Vogel approximation method-

From	A	B	C	Available
I.	2	7	4	5
II.	3	3	7	8
III.	5	4	1	7
IV.	1	6	2	14
Req.	7	9	18	

Q.2 Using the Simplex method, Solve the following LPP:

$$Z_{Max} = x_1 + 2x_2$$

$$\text{Sub. to } x_1 - x_2 \leq 10$$

$$3x_1 - 2x_2 \leq 40$$

$$\text{and } x_1, x_2 \geq 0$$

(CO5) L5

(CO5) L4

Q.3 Joint distribution function of two discrete random variable X and Y are given by $f(x, y) = c(2x + y)$ where x and y assume all integers such that $0 \leq x \leq 2, 0 \leq y \leq 3$. Then find (a) C, (b) $P(X=2, Y=1)$, (c) Marginal Distribution. (CO1) L5



GLOBAL INSTITUTE OF TECHNOLOGY

Roll No. 24E6JCS103

B. Tech. III Semester I-Midterm Exam 2025-26

3CS2-01/3AID2-01/3CS2-01/3CIT2-01/3CCS2-01

Branch-CS/AIDS/IT/IOT/CYBER SECURITY

Subject : Advanced Engineering Mathematics

29/10/2025/ Wednesday

Time: 2 Hours

Maximum Marks: 40

Attempt all questions
Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. no supplementary sheet shall be issued in any case.

MID-TERM-1

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1(Part-A)	5	2	1	1	1	10	10
CO2(Part-B)	1	3	1	0	0	5	15
CO2(Part-C)	0	0	1	1	1	3	15

Part A

(Answer should be given up to 25 words only)

All questions are compulsory

10x1 = 10

Q.1 What is Chebyshev's inequality? (CO1) L1

Q.2 Define the Spearman's formula for modified rank correlation coefficient for repeated rank. (CO1) L2

Q.3 Define moment generating function (MGF). (CO1) L2

Q.4 Define the Binomial distribution. (CO1) L1

Q.5 Write short note on history of optimization. (CO1) L1

Q.6 Define line of regression. (CO1) L5

Q.7A Variable X has the following probability mass function-

X	0	1	2	3	4	5	6	7
P(X)	0	K	2K	2K	3K	K ²	2K ²	7K ² + K

Find the value of K. (CO1) L1

Q.8 The joint probability density function of (X, Y) is given by:

$$f(x) = \begin{cases} \lambda xy, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Where λ is a constant. Find λ . (CO1) L3

Q.9 What is the variance of the Poisson distribution with mean value 5? (CO1) L1

Q.10 How optimization problems are classified based on the nature of expressions? (CO1) L4

Part B

Analytical/Problem solving questions

Attempt all questions (word Limit 75)

5 x 3 = 15

(CO2)

Q.1 Define Poisson Distribution. Derive it as a limiting case of Binomial distribution.

L1

Q.2 Write four engineering applications of optimization.

(CO2) L2

Q.3 Find the correlation coefficient between x and y when it is given that

$$n = 15, \sum x = 50, \sum y = -30, \sum x^2 = 290, \sum y^2 = 300, \sum xy = -115$$

(CO2) L2

Q.4 In a sample of 10 pens, if 10% pens produced by a company are defective. Find out the probability

(a) one pen must be defective

(b) at least two pens must be defective (CO2) L2

Q.5 Joint distribution function of two discrete random variables X and Y are given by

$f(x, y) = c(2x + y)$, where x and y assume all integer values such that $0 \leq x \leq 2, 0 \leq y \leq 3$. Find:

i) c

ii) $P(X = 2, Y = 1)$

iii) $P(X \geq 1, Y \leq 2)$

(CO2) L3

Part C

(Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

3 x 5 = 15

Q.1 Fit a straight line to the following data regarding x as an independent variable:

X	0	1	2	3	4
P(X)	1.0	1.8	3.3	4.5	6.3

(CO2) L4

Q.2 Write a short note on classification of optimization problems based on various parameter. (CO2) L3

Q.3 Calculate the first four moments about mean for the following distribution. (CO2) L5

X	0	1	2	3	4	5	6	7	8
Y	1	8	28	56	70	56	28	8	1

Roll No. 24E07CR103**GLOBAL INSTITUTE OF TECHNOLOGY****B. Tech. III Semester I-Midterm Exam 2025-26****3CS3-04/Digital Electronics****Branch-CS/AI/IT/CC/IOT**

30/10/2025/ Thursday

Maximum Marks: 40**Time: 2 Hours****Attempt all questions**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. no supplementary sheet shall be issued in any case.

MID-TERM-1

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1(Part-A)	3	4	2	1	0	10	10
CO2(Part-B)	0	3	1	1	0	5	15
CO2(Part-C)	0	0	1	2	0	3	15

Part A (Answer should be given up to 25 words only)
All questions are compulsory

- Q.1 Draw AND gate using NOR gate.
 Q.2 Draw XOR gate diagram and its truth table.
 Q.3 Convert $(10101101)_2$ to Gray?
 Q.4 Convert $(AB6)_{16}$ to $()_{10}$ $()_2$?
 Q.5 Convert $(102.57)_8$ into $(?)_{10}$.
 Q.6 Write expression of De Morgan's Theorem.
 Q.7 What is Minterm and Maxterm?
 Q.8 Define Sum of Product (SOP).
 Q.9 Define Product of Sum (POS).
 Q.10 $(245)_x = (131)_{10}$ Find the Value of x.

(CO1) L1
 (CO1) L1
 (CO1) L3
 (CO1) L3
 (CO1) L3
 (CO2) L1
 (CO2) L2
 (CO2) L2
 (CO2) L2
 (CO1) L4
10x1 = 10

Part B Analytical/Problem solving questions
Attempt all questions (word Limit 75)

- Q.1. Realize the Logic expression $Y = (A+B)(A'+C)(B+D)$ Using basic gate?
 Q.2 Subtract 748 from 983 using 9's complement method?
 Q.3. If $A = 1001$ and $B = 0100$ then find out the value of $A+B$ using BCD addition?
 Q.4 Prove That: $ab' + b'c' + a'c' = ab' + a'c'$.
 Q.5 Minimize the Given Boolean Function. $AB' + A'BC + ABC' + AB'C + ABC$

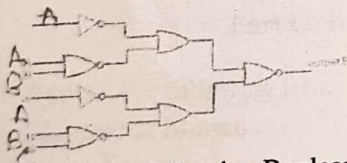
(CO1) L2
 (CO1) L3
 (CO1) L4
 (CO2) L2
 (CO2) L2

5 x3 =15

Part C (Descriptive/Analytical/Problem Solving/Design Question)
Attempt all questions (word Limit 125)

- Q.1. Draw the simplest possible logic diagram that implements the output of the logic diagram shown below.

(CO2) L4



- Q.2 Simplify the following Boolean function and obtain (i) minimal SOP (ii) minimal POS

(CO2) L3

$$F = \sum(0,2,3,6,7) + d(8,10,11,15)$$

- Q.3 Simplify the following Boolean function, $f(W, X, Y, Z) = \sum m(2,6,8,9,10,11,14,15)$ using Quine-McCluskey tabular method.

(CO2) L4

3x 5 =15

Roll No. 24EGJC5103**GLOBAL INSTITUTE OF TECHNOLOGY****B. Tech. III Semester II-Midterm Exam 2025-26****3CS3-04/Digital Electronics****Branch-CS/AI/IT/CC/IOT**

19/12/2025/ Friday

Maximum Marks: 40**Time: 2 Hours****Attempt all questions**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. no supplementary sheet shall be issued in any case.

MID-TERM-1

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1	2	4				6	8
CO2		1	1			2	4
CO3		1	2			3	9
CO4		1	2			3	11
CO5	2		2			4	8

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

Q.1 List the different types of flip flops.

(CO5) L1

Q.2 Define reflective codes.

(CO1) L1

Q.3 State De Morgan's theorem.

(CO1) L1

Q.4 Convert $(10101101)_2 \rightarrow ()_8$

(CO1) L2

Q.5 Explain race-around condition in JK flip flop.

(CO5) L3

Q.6 Illustrate Excitation table of SR flip flop.

(CO5) L1

Q.7 Explain don't care condition.

(CO2) L3

Q.8 Show the classification of digital logic families.

(CO3) L2

Q.9 Solve $(0100\ 1000.01111001)_{\text{Ex-3}} = ()_{10}$

(CO1) L2

Q.10 Calculate the value of x. $(23)_x + (12)_x = (101)_x$

(CO1) L2

10x1 = 10**Part B Analytical/Problem solving questions****Attempt all questions (word Limit 75)**

Q.1. What is multiplexer? Design 4:1 MUX using 2:1 MUX.

(CO4) L2

Q.2 Construct CMOS NAND and CMOS NOR gate for two inputs.

(CO3) L3

Q.3. Consider two binary numbers $X = 1010100$ and $Y = 1000011$, perform the subtraction using 2's complement.i) $X - Y$ ii) $Y - X$

(CO1) L2

Q.4 Design full adder circuit using half adders.

(CO4) L3

Q.5 Interpret the function $f = A + BC$ in canonical POS form (Product of Sum form).

(CO2) L2

5 x3 =15**Part C (Descriptive/Analytical/Problem Solving/Design Question)****Attempt all questions (word Limit 125)**

Q.1. Explain the following terms:

i) Noise Margin

ii) Propagation Delay

iii) Fan-In

iv) Fan-out

v) Power Dissipation

(CO3) L3

Q.2 Explain the working of 4-bit serial-in parallel-out shift register along with the waveform.

(CO5) L3

Q.3 Design a 4-bit binary to gray code converter and realize it using logic gates.

(CO4) L3

3x 5 =15



GLOBAL INSTITUTE OF TECHNOLOGY

B. Tech. III Semester II-Midterm Exam 2025-26

JCS4-07: Software Engineering

Branch-CS/IT/AIDS/IOT/Cyber Security

19/12/2025/ Friday

Time: 2 Hours

Maximum Marks: 40

Attempt all questions
Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1(Part A, B, C)	3	1	0	0	0	4	6
CO2 (Part A, B, C)	0	0	1	0	2	3	9
CO3 (Part A, B)	0	1	1	2	0	4	6
CO4 (Part A, B, C)	0	1	2	0	0	3	9
CO5 (A, B, C)	0	2	0	2	0	4	10

Part A (Answer should be given up to 25 words only)
All questions are compulsory

- Q.1 Define software. Enlist the characteristics of good software.
Q.2 Give the difference b/w FP and LOC.
Q.3 What is SRS?
Q.4 Explain FSM model.
Q.5 Why accuracy is important attribute for a data dictionaries.
Q.6 What is software Design. Write any Four Design principles.
Q.7 What is Input /Process/Output (IPO) approach in S/W Design.
Q.8 What do you mean by OO concept. Write 3 OO principles.
Q.9 Explain the term Risk Analysis. Enlist Four major categories of Risk analysis.
Q.10 Differentiate b/w object-oriented analysis (OOA) and Object-Oriented Design (OOD).

CO1(L1)

CO2(L3)

CO1(L1)

CO3(L2)

CO3(L3)

CO4(L2)

CO1(L1)

CO5(L2)

CO3(L4)

CO5(L4)

10 x 1 = 10

5 x 3 = 15

CO5(L2)

CO3(L4)

CO4(L3)

CO1(L2)

CO2(L5)

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

- Q.1 Write a short note on Object Oriented Design concepts.
Q.2 Give the difference b/w DFD and CFD with proper example and diagram.
Q.3 What is a good Software Design? Explain the Design Documentation with example.
Q.4 What is prototyping? Give the sequence of events needed in prototyping.
Q.5 Suppose that a project was estimated to be 400 KLOC. Calculate effort and time for each of three modes of development.

Table given as:

Mode	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

- Q.1 Explain Effective modular design in terms of cohesion and coupling with all its types and diagram.
Q.2 Define the term UML. How it is useful in object-oriented modeling. Explain the following in context of UML
CO5(L4)
a) Use case diagram
b) State transition diagram.

3X5=15

CO4(L3)

Q.3. Compute the function point productivity, documentation, cost per function for the following data:

Measurement Parameter	Count	Weighting
No. of External Input	24	4
No. of External output(E0)	46	4
No. of External Inquiries (EC)	8	6
No. of Internal files (ILF)	4	10
No. of External interfaces (EIF)	2	5

CO2(L5)

iv) Effort - 36.9 PM

v) Technical documents - 265 inn

vi) User documents 122 pages

vii) Cost - \$ 7744/month

Various processing factors are: 4, 1, 0, 3, 3, 5, 4, 4, 3, 3, 2, 2, 4, 5.



GLOBAL INSTITUTE OF TECHNOLOGY

B. Tech. III Semester I-Midterm Exam 2025-26

3CS4-07: Software Engineering

Branch-CS/IT/AIDS/IOT/Cyber Security

30/10/2025/ Thursday

Maximum Marks: 40

Time: 2 Hours

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1(Part-A,B,C)	2	3	3	1	1	10	18
CO2 (Part A,B,C)	1	2	1	0	2	6	16
CO3 (Part A,,C)	0	1	0	1	0	2	6

Part A (Answer should be given up to 25 words only)

All questions are compulsory

- Q.1 Define a software process model. (CO1) L1
 Q.2 Explain the need of software Engineering (CO1) L1
 Q.3 Define formal requirements specification? (CO1) L2
 Q.4 Differences between verification and validation. (CO1) L3
 Q.5 Describe the importance of documentation in software engineering? (CO1) L2
 Q.6 Explain Project Manager responsibilities. (CO2) L1
 Q.7 Difference between LOC and FP estimation. (CO2) L2
 Q.8 Explain requirement traceability. (CO1) L3
 Q.9 Mention any two advantages of the Prototype Model. (CO1) L2
 Q.10 What is the role of stakeholders in requirement analysis? (CO3) L2
10x1 = 10

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

5 x3 =15

- Q.1 Explain the quadrant of spiral model. (CO1) L3
 Q.2 Difference between functional and non-functional requirements? (CO1) L4
 Q.3 Describe phases of the software life cycle. (CO2) L2
 Q.4 Explain resource estimation in software project management. (CO2) L3
 Q.5 A software project content 12 modules. Each module is expected to have about 250 lines of code if a programmer writes 100 LOC per day calculate.
 a) Total Size of Project (LOC)
 b) Total effort in person days
 c) Productivity
30 marks
 (CO2) L5

Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

3X5=15

- Q.1 Discuss how to prepare a software Requirement specification (SRS) with its features? (CO1) L5
 Q.2 Explain type of requirement and requirement analysis principles in detail. (CO3) L4
 Q.3 Suppose a project was estimated to be 500 KLOC. Calculate the effort, development time and average resource size (average person required) for each of the three models Organic mode semi-detached, embedded. (CO2) L5

Software project	a	b	c	d
Organic	2.4	1.05	2.5	0.38
semidetached	3.0	1.02	2.5	0.35
embedded	3.6	1.20	2.5	0.32

Roll No. 24E6TCS103

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. III Semester I midterm Exam 2025-26
3CS4-05/Data Structures and Algorithms
CS/IT/AI&DS/IOT/Cyber Security

28/10/2025 & Tuesday

Time: 2 Hours**Maximum Marks: 40****Attempt all questions**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. **No supplementary sheet shall be issued in any case.**

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total No. of Questions	Total Marks
CO1(Part-A)	2	4	3	1	0	10	10
CO2(Part-B)	1	2	1	0	1	5	15
CO2(Part-C)	0	1	1	0	1	3	15

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

- Q.1 Define algorithm. (CO1) L1
Q.2 Explain the Header LinkedList. (CO1) L2
Q.3 What do you mean by priority queue. (CO1) L1
Q.4 Differentiate between static and Dynamic memory allocation. (CO1) L3
Q.5 Explain Stack operations. (CO1) L2
Q.6 What are the differences between normal queue and circular queue? (CO1) L3
Q.7 Discuss the linear and non-linear data structures. (CO1) L4
Q.8 Explain circular linked list. (CO1) L2
Q.9 What is difference between array & queue. (CO1) L3
Q.10 Explain the logic that checks whether a Queue is empty or not for a Queue implemented using array. (CO1) L2
10×1=10

Part B Analytical/Problem solving questions**Attempt all questions (word Limit 75)**

- Q.1 Write an algorithm to delete and insert a data item in circular queue. (CO2) L1
Q.2 Write down the algorithm to implement stack using Linked list. (CO2) L3
Q.3 Explain Asymptotic notations: Big-Oh, theta, Omega using suitable example. (CO2) L2
Q.4 Explain the tower of Hanoi problem in detail and write algorithm for that. (CO2) L2
Q.5 Convert following infix expression to postfix expression:
a. $A+B/C-D^E-F$
b. $A/B-(C+D) * E/F^G$ (CO2) L5
5×3=15

Part C (Descriptive/Analytical/Problem Solving/Design Question)**Attempt all questions (word Limit 125)**

- Q.1 (a) How to perform factorial calculation using stack? Explain. (CO2) L2
(b) Write algorithms steps to convert infix to prefix expression using stack.
Q.2 What is doubly linked list? Explain the algorithms for inserting a node and deleting a node from doubly linked list. (CO2) L3
Q.3 What is Dequeue? Write down the algorithms for insertion and deletion operations performed on Dequeue. (CO2) L5
3×5=15



Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Course Outcomes	Level-1	Level-2	Level-3	Level-4	Level-5	Total no of Questions	Total Marks
CO1(Part A)	0	1	0	0	0	1	1
CO2(Part A, B, C)	2	3	0	2	0	7	15
CO3(Part A, B)	2	0	0	1	0	3	5
CO4(Part A, C)	2	1	0	0	0	3	7
CO5(Part A, B, C)	1	1	0	2	0	4	12

Part A (Answer should be given up to 25 words only)

All questions are compulsory

- Q.1 How stack is represented using dynamic array? Explain with example. (CO2, L4)
 Q.2 Identify the applications of stack. (CO2, L1)
 Q.3 Elaborate the disadvantages of Linked List over Arrays. (CO5, L1)
 Q.4 Justify how circular queue is different from normal queue. (CO2, L1)
 Q.5 Differentiate between sequential and binary search. (CO1, L2)
 Q.6 Illustrate different traversal techniques used in binary search tree. (CO3, L1)
 Q.7 List out best, average and worst-case complexity of merge and quick sort. (CO3, L1)
 Q.8 Examine the relationship between number of nodes and height of AVL. (CO4, L1)
 Q.9 Determine advantages and disadvantages of adjacent matrix representation for graphs. (CO2, L2)
 Q.10 Explain how quadratic probing is done. (CO4, L1)

10 × 1 = 10

Part B (Analytical/Problem solving questions)

Attempt all questions (word Limit 75)

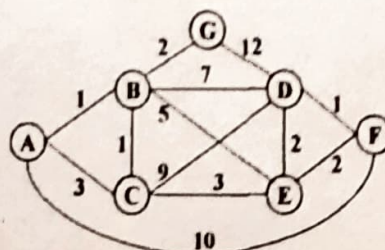
- Q.1 Convert the following infix expression into postfix expression: (CO5, L4)
 $A + B - (C + D) / E * F - (G + H) / I$
 Q.2 Consider the preorder of a BST: (CO5, L4)
 Pre-order: 20, 5, 3, 4, 10, 15, 30, 25, 40
 What will be the post-order?
 Q.3 Sort the following elements using quick sort: (CO2, L2)
 28, 5, 16, 36, 11, 19, 25
 Q.4 Write an algorithm for DFS traversal. (CO2, L2)
 Q.5 What is a MST? Differentiate between Kruskal and Prim's algorithm with their time complexity. (CO3, L4)

5 × 3 = 15

Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

- Q.1 Examine each step involved with Radix sort to sort the given array – (CO2, L4)
 329, 457, 657, 839, 436, 720, 355
 Q.2 Create an AVL tree using the following nodes 10, 6, 11, 12, 1, 7, 0, 2, 3. What will be the resulting AVL tree if node 12 is deleted from above AVL tree? (CO4, L2)
 Q.3 Consider the following graph - (CO5, L2)



Find the minimum spanning tree using Prim's algorithm.

3 × 5 = 15